

Course Outcomes:

- The student will learn core data types of python.
- The student will learn conditional and looping operations in python.
- The student will be able to work with Object-oriented concepts and Database connectivity in python.
- The student will be able to analyze data using Pandas and Numpy.
- The student will be able to visualize the data using seaborn and matplotlib.

DS7A-711 NoSQL Laboratory**Credits: 3 (1-0-4)**

Objective: This laboratory course will discuss the NoSQL databases and the characteristics that distinguish them from traditional relational database management systems. Core concepts of NoSQL databases will be presented, followed by an exploration of how different database technologies implement these core concepts. Students will engage in practical exercises that involve 1-2 databases from each of the four main NoSQL data models (key-value, column family, document, and graph), highlighting the business needs that drive the development and use of each database. By the end of the course, students will have gained hands-on experience with various NoSQL databases and will be able to make informed decisions about database technologies for specific business applications.

COURSE DESCRIPTION:

UNIT I: Introduction to NoSQL: Basics, Storage Architecture, Operations, Query Model, Modifying Data Stores and Managing Evolution, Indexing and Ordering Data Sets, Managing Transactions and Data Integrity.

UNIT II: Introduction to Key-Value Databases (Redis Lab), Document Stores (MongoDB Lab), Column Family Stores (Cassandra Lab), Graph Databases (Neo4j Lab). Choosing a NoSQL Database.

UNIT III: MongoDB functions, operators and types of operations in MongoDB, use of Indexing, Advanced Indexing, aggregation and Map Reduction.

UNIT IV: Make use of MongoDB commands and queries, aggregate pipelines to extract data, optimization of queries by creating indexes, Development of aggregate pipelines for text search in collections

Text Books:

1. Harrison, G., Next-Generation Databases. Apress. ISBN: 9781484213292.
2. Sullivan, D., NoSQL for Mere Mortals. Addison-Wesley Professional. ISBN: 0134023218.
3. Chodorow, K., MongoDB: The Definitive Guide (2nd ed.). O'Reilly Media.
4. Banker, K., et al., MongoDB in Action (2nd ed.). Manning Publications.

5. Sharma, M., MongoDB Complete Guide (1st ed.). BPB Publications.
6. Installation of MongoDB: <https://www.youtube.com/watch?v=dEm2AS5amyA>
7. Aggregation: <https://www.youtube.com/watch?v=vx1C8EyTa7Y>
8. MongoDB in Action: <https://www.manning.com/downloads/529>
9. MongoDB Exercises: <https://www.w3resource.com/mongodb-exercises/>

Suggested List of Programs to be Discussed in Lab:

1. Retrieve all movies from the 'movies' collection that have complete information and have received more than 1000 votes on IMDb.
2. Retrieve all movies from the 'movies' collection that have complete information and have an IMDb rating higher than 7.
3. Retrieve all movies from the 'movies' collection that have complete information and have a viewer rating higher than 4 on Tomatoes.
4. Retrieve all movies from the 'movies' collection that have received an award.
5. Find all movies with title, languages, released, runtime, directors, writers, countries from the 'movies' collection in MongoDB that have a runtime between 60 and 90 minutes.
6. Find all movies with title, languages, released, runtime, directors, writers, countries, imdb from the 'movies' collection in MongoDB for the top 5 movies with the highest IMDb ratings.
7. Find all movies from the 'movies' collection in MongoDB with the average runtime of movies released in each country.
8. Find from the 'movies' collection in MongoDB with the most common genre among the movies.

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9. Find the movies released in the year with the highest average IMDb rating from the 'movies' collection in MongoDB.
10. Find the top 10 directors with the most movies from the 'movies' collection in MongoDB.
11. Write a query in MongoDB to find the average IMDb rating for movies with different ratings (e.g., 'PG', 'R', 'G') from the 'movies' collection.
12. Write a query in MongoDB to find the oldest movie with an award win from the 'movies' collection.
13. Write a query in MongoDB to find the movie with the highest IMDb rating and viewer rating on Tomatoes from the 'movies' collection.

14. Write a MongoDB query to find the restaurants that do not prepare any cuisine of 'American' and their grade score more than 70 and latitude less than -65.754168.
15. Write a MongoDB query to find the restaurants which do not prepare any cuisine of 'American' and achieved a score more than 70 and located in the longitude less than -65.754168. Note : Do this query without using \$and operator.
16. Write a MongoDB query to find the restaurants which do not prepare any cuisine of 'American' and achieved a grade point 'A' not belongs to the borough Brooklyn. The document must be displayed according to the cuisine in descending order.
17. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which contain 'Wil' as first three letters for its name.
18. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which contain 'ces' as last three letters for its name.
19. Find all listings with listing_url, name, amenities, host in the listingsAndReviews collection that have a host with a Jumio verification and a about section.
20. Retrieve all documents with listing_url, name, host, price in the listingsAndReviews collection where the host_total_listings_count field is greater than 1.
21. Retrieve all documents with listing_url, name, property_type, bed, price in the listingsAndReviews collection where the property_type field is equal to "Apartment" and the beds field is greater than or equal to 2.
22. Find all listings with listing_url, name, property_type, bed, bathrooms, price in the listingsAndReviews collection that have a minimum of 2 bathrooms.
23. Find all listings with listing_url, name, property_type, bed, price, guests_included in the listingsAndReviews collection that have a maximum of 5 guests included in the price.

DS7A-713: Advanced Excel

Credits: 3 (1-0-4)

Objective: The main objective of this course is to learn analysis of data using MS Excel, resulting in less time and better understands what the data means.

COURSE DESCRIPTION:

Unit I: Introduction to Excel User Interface, Application, Workbook, Worksheets & its Components, Named Ranges; Formatting: Cell Color, Font Color, Indents, Alignments, Number Formats, Custom Formats, Editing commands; Data Sorting: Built-in Sort, Sorting Levels, Custom Sort; Data Filtering: Auto Filter – Filter By Color, Filter by Icon, Advanced Filter, Remove Duplicates; Data Subtotal – Built-In Subtotal (Nested Subtotal).